Tejaswi Venumadhav Nerella

Compact Curriculum Vitae

Associate Professor, Department of Physics, University of California, Santa Barbara
Broida Hall, Santa Barbara, CA 93106-9530 teja@ucsb.edu +1 (626) 826-3571

Appointments and Education

University of California, Santa Barbara Jul 2026–present
Associate Professor, Department of Physics

University of California, Santa Barbara Jul 2020–Jul 2026
Assistant Professor, Department of Physics

International Center for Theoretical Sciences, Bangalore 2020–present
Visiting Professor

Institute for Advanced Study, Princeton 2015–2020
Member; Schmidt Fellow and John Bahcall Fellow

California Institute of Technology 2010–2015
Ph.D. in Physics; advisor: Christopher M. Hirata

Indian Institute of Technology, Kanpur 2005–2010
Integrated M.Sc. in Physics

Research Profile

- Theoretical astrophysics and gravitational-wave data analysis, with emphasis on compact-binary searches, parameter estimation, waveform modeling, strong lensing, neutron-star physics, and early-universe cosmology.
- Developer and user of large-scale computational methods for gravitational-wave inference and searches in non-Gaussian, non-stationary detector data.
- Publications: 50 refereed papers, 22 preprints, 3 large-collaboration papers; total h-index 32 from ADS-backed publication data.

Computational Methods and Codes

- Co-developer of a full gravitational-wave compact-binary search code for LIGO/Virgo data, including matched filtering, template-bank methods, ranking statistics, background estimation, and follow-up of candidates in non-Gaussian detector noise.
- Developer of *cogwheel*, a gravitational-wave parameter-estimation code for compact-binary inference, designed for efficient likelihood evaluation, sampling, and post-processing of detector data.
- Developer of additional open research codes for cosmological thermal history, sterile-neutrino dark matter, early-universe recombination physics, and related gravitational-wave/cosmology calculations.
- Experience designing large ensembles of independent analyses, validating scientific workflows across heterogeneous compute resources, and turning production-scale runs into reproducible publications and public data products.

Selected Funding, Honors, and Awards

- NSF, *WoU-MMA: Targeted Search for Binary Mergers with Multiple Harmonics in Gravitational Wave Data*, PI, 2023–2026.
- NSF, *WoU-MMA: Expanding the Horizons of Gravitational Wave Searches and Parameter Estimation*, PI, 2020–2024.
- Alfred P. Sloan Research Fellowship, 2023; Hellman Family Faculty Fellowship, 2023.
- John Bahcall Fellowship, Institute for Advanced Study, 2019; Schmidt Fellowship, Institute for Advanced Study, 2015–2018.
- International Fulbright Science and Technology Award, 2010; President's Gold Medal, IIT Kanpur, 2010.

Mentoring, Leadership, and Service

- Research mentor to graduate students, postdoctoral scholars, undergraduate researchers, and KITP fellows in gravitational-wave astrophysics, compact-object inference, waveform modeling, and cosmology; current group projects include search pipelines, higher harmonics, eccentricity, neutron-star tides, and fast inference.
- Organizer: Prospects in Theoretical Physics 2025 program on *Gravitational Waves from Theory to Observation*; KITP 2025 program *Stellar-Mass Black Holes at the Nexus of Optical, X-ray, and Gravitational Wave Surveys*; KITP 2025 conference *The Lifecycle of Stellar Black Holes*.
- Referee for Astrophysical Journal, Astrophysical Journal Letters, Monthly Notices of the Royal Astronomical Society, Astroparticle Physics, and Physical Review D; panel reviewer for ERC, BSF, and NSF.

Selected Publications

1. Hegade K. R., A., Kwon, K.J., **Venumadhav, T.**, Yu, H., Yunes, N. (2026). *Relativistic and Dynamical Love Numbers.* Physical Review Letters, 136, 071401.
2. Cheung, M.H.Y., Wadekar, D., Mehta, A.K., Islam, T., Roulet, J., Berti, E., **Venumadhav, T.**, et al. (2026). *Searching for intermediate mass ratio binary black hole mergers in the third observing run of LIGO-Virgo-KAGRA.* Physical Review D, 113, 023003.
3. Mushkin, J., Roulet, J., Zackay, B., **Venumadhav, T.**, Ivashtenko, O., Wadekar, D., Mehta, A.K., Zaldarriaga, M. (2025). *Sampler-free gravitational wave inference using matrix multiplication.* Physical Review D, 112, 104025.
4. Islam, T., **Venumadhav, T.**, Mehta, A.K., Anantpurkar, I., Wadekar, D., Roulet, J., Mushkin, J., et al. (2025). *Data-driven extraction, phenomenology, and modeling of eccentric harmonics in binary black hole merger waveforms.* Physical Review D, 112, 044070.
5. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M. (2020). *New binary black hole mergers in the second observing run of Advanced LIGO and Advanced Virgo.* Physical Review D, 101, 083030.
6. Olsen, S., **Venumadhav, T.**, Mushkin, J., Roulet, J., Zackay, B., Zaldarriaga, M. (2022). *New binary black hole mergers in the LIGO-Virgo O3a data.* Physical Review D, 106, 043009.
7. Zackay, B., **Venumadhav, T.**, Dai, L., Roulet, J., Zaldarriaga, M. (2019). *Highly spinning and aligned binary black hole merger in the Advanced LIGO first observing run.* Physical Review D, 100, 023007.
8. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M. (2019). *New search pipeline for compact binary mergers: Results for binary black holes in the first observing run of Advanced LIGO.* Physical Review D, 100, 023011.
9. Zackay, B., Dai, L., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M. (2021). *Detecting gravitational waves with disparate detector responses: Two new binary black hole mergers.* Physical Review D, 104, 063030.
10. Roulet, J., Chia, H.S., Olsen, S., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M. (2021). *Distribution of effective spins and masses of binary black holes from the LIGO and Virgo O1-O3a observing runs.* Physical Review D, 104, 083010.
11. **Venumadhav, T.**, Cyr-Racine, F.Y., Abazajian, K.N., Hirata, C.M. (2016). *Sterile neutrino dark matter: Weak interactions in the strong coupling epoch.* Physical Review D, 94, 043515.
12. Dai, L., **Venumadhav, T.**, Sigurdson, K. (2017). *Effect of lensing magnification on the apparent distribution of black hole mergers.* Physical Review D, 95, 044011.
13. Zackay, B., **Venumadhav, T.**, Roulet, J., Dai, L., Zaldarriaga, M. (2021). *Detecting gravitational waves in data with non-stationary and non-Gaussian noise.* Physical Review D, 104, 063034.
14. **Venumadhav, T.**, Dai, L., Miralda-Escudé, J. (2017). *Microlensing of Extremely Magnified Stars near Caustics of Galaxy Clusters.* The Astrophysical Journal, 850, 49.
15. Mehta, A.K., Olsen, S., Wadekar, D., Roulet, J., **Venumadhav, T.**, Mushkin, J., Zackay, B., Zaldarriaga, M. (2025). *New binary black hole mergers in the LIGO-Virgo O3b data.* Physical Review D, 111, 024049.

16. **Venumadhav, T.**, Dai, L., Kaurov, A., Zaldarriaga, M. (2018). *Heating of the intergalactic medium by the cosmic microwave background during cosmic dawn.* Physical Review D, 98, 103513.
17. Roulet, J., **Venumadhav, T.**, Zackay, B., Dai, L., Zaldarriaga, M. (2020). *Binary black hole mergers from LIGO/Virgo O1 and O2: Population inference combining confident and marginal events.* Physical Review D, 102, 123022.
18. Kaurov, A.A., Dai, L., **Venumadhav, T.**, Miralda-Escudé, J., Frye, B. (2019). *Highly Magnified Stars in Lensing Clusters: New Evidence in a Galaxy Lensed by MACS J0416.1-2403.* The Astrophysical Journal, 880, 58.
19. Dai, L., **Venumadhav, T.**, Kaurov, A.A., Miralda-Escud, J. (2018). *Probing Dark Matter Subhalos in Galaxy Clusters Using Highly Magnified Stars.* The Astrophysical Journal, 867, 24.
20. Roulet, J., Olsen, S., Mushkin, J., Islam, T., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M. (2022). *Removing degeneracy and multimodality in gravitational wave source parameters.* Physical Review D, 106, 123015.

Selected Recent Preprints

1. Ho-Yeuk Cheung, M., Wadekar, D., Zaldarriaga, M., **Venumadhav, T.**, Roulet, J., Mehta, A.K. (2026). *The diffraction-lensing interpretation of GW231123 with astrophysical priors.* arXiv:2607.21834.
2. Yu, H., Nicolini, G., Lau, S.Y., Kwon, K.J., **Venumadhav, T.**, Andersson, N., Pnigouras, P., et al. (2026). *Nonlinear hydrodynamics in spinning neutron stars: Theoretical universal relations and equilibrium solutions.* arXiv:2607.07943.
3. Abhishek Hegade K., R., Kwon, K.J., **Venumadhav, T.**, Yu, H., Yunes, N. (2026). *The Good, the Bad, and the Subtle: Relativistic mode sums for neutron-star tidal response.* arXiv:2605.08569.
4. Perera, A., Zackay, B., **Venumadhav, T.** (2026). *Expanding the Population of Short Gamma-Ray Transients with a Coherent Fermi/GBM Search. A 13-year catalog of short GRBs.* arXiv:2605.31554.

Full publication list: `https://ui.adsabs.harvard.edu/search/q=author%3A%22Venumadhav%2C%20Tejaswi%22&sort=date%20desc%2C%20bibcode%20desc&p_=0`